Note 1 - Functions & Domains

Function Definitions

- **Function**: A rule or law that governs the relationship between two variables in which the values of one variable depend on the names of the other variable.

  $$y = f(x)$$

  In the above equation, x is the independent variable and $f(x)$ is the dependent variable.

- **Domain**: The set of all real numbers (inputs), x, for which the function is defined or exists.

- **Range**: The set of all real numbers (outputs), $f(x)/y$, for which the function is defined or exists.
  Note: The variable names x, $f(x)$, and y are standard, but are also arbitrary.

- **Piecewise function**: A function that is defined by different formulas in different parts of the domain.

  For example, the absolute value function is defined as

  $$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

- **Even function**: A function that satisfies $f(-x) = f(x)$ for every number x in its domain.
  For example,

  $$\begin{aligned} f(-x) &= (-x)^2 \\ &= x^2 \\ &= f(x) \end{aligned}$$

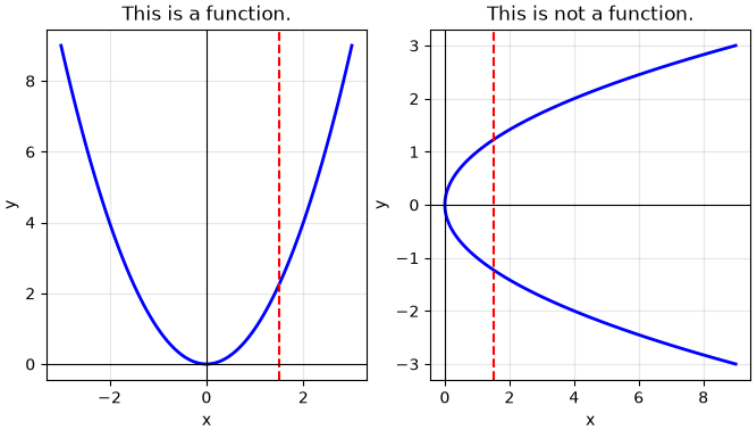
- **Odd function**: A function that satisfies $f(-x) = -f(x)$ for every number x in its domain.
  For example,

  $$\begin{aligned} f(-x) &= (-x)^3 \\ &= -x^3 \\ &= -f(x) \end{aligned}$$

- **Increasing function**: A function is increasing on an interval if $f(x_1) < f(x_2)$ when $x_1 < x_2$ on the interval.

- **Decreasing function**: A function is decreasing on an interval if $f(x_1) > f(x_2)$ when $x_1 < x_2$ on the interval.

- **Vertical line test**: A curve in the xy-plane is the graph of a function of x if and only if no vertical line passes through the curve more than once.
  For example, the vertical line test tells us that for $y = x^2$ and $x = y^2$:



Methods for Finding the Domain

Method 1: Fractional (Rational) Functions
Procedure: Find values of x that make the denominator 0.

- Set denominator equal to 0 and solve for x.
- The domain is the set of all real numbers except the values of x found in previous step.
  Recall: A denominator of 0 makes the function undefined.

Example: $h(x) = \dfrac{3-x}{2x+1}$

$$\begin{aligned} 2x+1 &= 0 && \text{Set the denominator equal to 0.} \\ 2x &= -1 && \text{Solve for x.} \\ x &= -\frac{1}{2} \end{aligned}$$

Thus, the domain is everywhere $x \neq -\frac{1}{2}$, since plugging in $-\frac{1}{2}$ would lead to invalid outputs.
In interval notation, this is written as

$$(-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$$

Note: The parentheses are used to show that the interval is not inclusive of $-\frac{1}{2}$ in the domain, but includes numbers greater than $-0.4\overline{999}$ and less than $-0.5000...1$. The $\cup$ represents the union of two subdomains.

Method 2: Even Root Functions *i.e.* $\sqrt{x}, \sqrt[4]{x}, \sqrt[6]{x}...$
Procedure: Find values of x that make the expression inside the root greater than or equal to 0.

- Set the expression inside the radical it greater than or equal to 0, then solve the polynomial inequality.
- The solution to the inequality is the domain.
  Recall: Negative numbers inside even roots lead to imaginary solutions.

Example: $g(x) = \sqrt{x^2 - x - 6}$

$$\begin{aligned} x^2 - x - 6 &\geq 0 && \text{Set the expression greater than or equal to 0.} \\ (x-3)(x+2) &= 0 && \text{Solve for } x\text{. } \textit{Note: the inequality is temporarily changed to an equality.} \\ x &= 3, -2 && \text{Thus, the inequality is equal to 0 when } x = 3 \text{ or } -2. \end{aligned}$$

When $x = 3$ or -2, then we know that the inequality is equal to 0. However, we must also check whether the inequality is true for the intervals where $x < -2, -2 < x < 3$, and $x > 3$. For each interval, plug in a test point for x and see if the inequality holds true.

- For $x < -2$: plugging in $x = -3$ gives $(-3)^2 + 3 - 6 \geq 0$. When simplified, this gives $6 \geq 0$, which is true. Thus, real numbers less than or equal to -2 are in the domain.
- For $-2 < x < 3$: plugging in $x = 0$ gives $(0)^2 - 0 - 6 \geq 0$. When simplified, this gives $-6 \geq 0$, which is false. Thus, x-values between -2 and 3 are *not* in the domain of this function.
- For $x > 3$: plugging in $x = 4$ gives $(4)^2 - 4 - 6 \geq 0$. When simplified, this gives $6 \geq 0$, which is true.

In interval notation, this is written as

$$(-\infty, -2] \cup [3, \infty)$$

Note: The brackets are used to show that the interval is inclusive of -2 and 3 in the domain.